

Samanwaya Datta

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Summary

Computer Science student (VIT Chennai, 2027) with hands-on experience across backend systems and applied machine learning, including a distributed Go-based job queue and a full-stack agentic RAG application with conversational memory and retrieval routing. Strong foundation in DSA and core ML algorithms, with proven ability to design and ship complete systems independently.

Education

Vellore Institute of Technology, Chennai

Aug 2023 – Present

B.Tech in Computer Science Engineering (Electrical & CS) — CGPA: 8.83 / 10.0 (88.3%)

St. Patrick's Higher Secondary School, Asansol

2010 – 2023

Class X: 95% — Class XII: 93%

Projects

RAG Study Assistant – Agentic AI Application

[Source Code](#)

- Built an AI study assistant that reads uploaded PDFs and notes, then answers questions about them in a natural, multi-turn conversation that remembers earlier context.
- Made the system intelligently decide **when to re-search documents versus continue the conversation from memory**, cutting unnecessary lookups and speeding up responses.
- Designed the document-search engine to be swappable between two backends (**FAISS, ChromaDB**) with a single config change; packaged the full app with **Docker** and a **React** drag-and-drop interface.
- **Tech Stack:** Python, FastAPI, Gemini API, FAISS, ChromaDB, React, Docker

NotifyQ – Distributed Email Queue System

[Source Code](#)

- Built a notification scheduling system in **Go** that reliably delivers emails at the right time, handling thousands of delayed jobs without losing any.
- Split the system into **2 independent services** (one that accepts requests, one that processes them) so the app stays fast and responsive even while clearing a large backlog.
- Designed the system so failed emails **automatically retry with increasing wait times** and get logged separately after repeated failures instead of silently disappearing; built **5+ REST endpoints** with job data persisted in **Redis**.
- **Tech Stack:** Go, Redis, REST APIs, SMTP, Docker

Algorithmic Profile

- Solved **128+ problems** on LeetCode leetcode.com/u/Sammie156
- Strong focus on **Dynamic Programming, Backtracking, and Trees**; consistent practice across Arrays, Strings, Two Pointers, and Graph traversal patterns.

Technical Skills

- **Languages:** Go, Python, JavaScript, C++, Java, C
- **Machine Learning:** Scikit-learn, Regression, Decision Trees, Random Forest, XGBoost, SVM, KNN, K-Means, Neural Networks (PyTorch), Pandas, NumPy
- **AI / LLM Systems:** RAG, Agentic Routing, Vector Embeddings, FAISS, ChromaDB, Google Gemini API, Sentence-Transformers
- **Backend & Systems:** FastAPI, Go, Node.js, Express.js, REST APIs, JWT, Redis, Goroutines, Background Job Processing
- **Databases:** MySQL, MongoDB, Redis, SQL
- **Frontend:** React, TailwindCSS
- **Tools:** Git, Docker, Postman, Linux, AWS (EC2, S3)
- **Core CS:** Data Structures & Algorithms, OOP, System Design, OS, Computer Networks

Experience

JavaScript Intern | *Atyantik Technologies*

May 2025 – Jul 2025

- Integrated RESTful APIs into client-facing features, reducing average data fetch latency by **20%** and improving overall user experience across the application.
- Engineered reusable React component library and optimized Redux state management, improving codebase scalability and reducing redundant code by approximately 30%.
- Collaborated in two-week Agile sprints with senior developers; maintained clean commit history through Git branching, pull requests, and structured code reviews on GitHub.

Awards & Leadership

- **IoT Finalist** – Technex, IIT BHU; Top 10 National Rank for designing a hardware-software integrated agriculture system. 2025
- **All-Rounder Excellence Award** – St. Patrick's Higher Secondary School, Asansol May 2023